

Non-Calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

You have 30 minutes.

Signature _____

SHOW ALL WORK FOR FULL CREDIT!

Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. (More than one of these descriptions may apply.) (2 points)

1.

a) $-\sqrt{25}$

b) $2i\sqrt{3}$

Find the quotient. Write your answer in standard form. (2 points)

2. $\frac{\sqrt{-54}}{\sqrt{6}}$

Simplify each power of i . (1 point each)

3.

a) i^{30}

b) $\frac{1}{i^9}$

Is the given number a solution of the equation? Show work. (2 points)

4. $3 - 2i$; $x^2 - 6x + 13 = 0$

YES / NO

Find the product. Write your answer in standard form. (2 points)

5. $(2 - 7i)^2$

Find the quotient. Write your answer in standard form. (3 points)

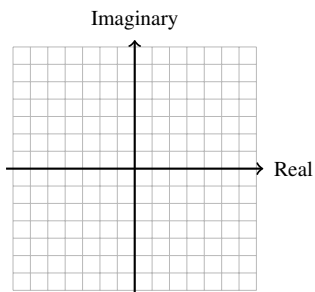
6. $\frac{5 - i}{3 + 2i}$

Solve the equation over the set of complex numbers. (4 points)

7. $3x^2 + 5 = 2x$

Find the sum of the pair of complex numbers. Write your answer in standard form. Graph both complex numbers and their resultant. Then find the absolute value of the sum. (3 points)

8. $-4 + i$, $1 - 5i$



Write the complex number in rectangular form. (3 points)

9. $4 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$

Write each complex number in trigonometric (polar) form, with θ in the interval $[0^\circ, 360^\circ)$. (2 points each)

10.

a) $1 - i\sqrt{3}$

b) $-7i$

The Julia set: compute $z^2 - 1$, $(z^2 - 1)^2 - 1$, $[(z^2 - 1)^2 - 1]^2 - 1$, ... If the absolute value of any result exceeds 2, z is **not** in the Julia set; otherwise it is. (3 points)

11. Is $z = \frac{1}{2}i$ in the Julia set? Show work.

YES / NO

ANSWER KEY – Non-Calculator**SHOW ALL WORK FOR FULL CREDIT!**

Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. (More than one of these descriptions may apply.) (2 points)

1.

a) $-\sqrt{25}$

b) $2i\sqrt{3}$

a) -5 ; $b = 0$.

b) $a = 0$, $b = 2\sqrt{3} \neq 0$.

complex, real

complex, pure imaginary, nonreal complex

Find the quotient. Write your answer in standard form. (2 points)

2. $\frac{\sqrt{-54}}{\sqrt{6}}$

$\sqrt{-54} = i\sqrt{54} = 3i\sqrt{6}$ $\frac{3i\sqrt{6}}{\sqrt{6}} = \boxed{3i}$

Simplify each power of i . (1 point each)

3.

a) i^{30}

b) $\frac{1}{i^9}$

a) $30 \div 4 = 7 \text{ R } 2$, so $i^{30} = i^2 = \boxed{-1}$

b) $i^9 = i$; $\frac{1}{i} \cdot \frac{-i}{-i} = \frac{-i}{1} = \boxed{-i}$

Is the given number a solution of the equation? Show work. (2 points)

4. $3 - 2i$; $x^2 - 6x + 13 = 0$

$(3 - 2i)^2 = 9 - 12i + 4i^2 = 5 - 12i$ $-6(3 - 2i) = -18 + 12i$

$(5 - 12i) + (-18 + 12i) + 13 = 0 \checkmark$

YES

Find the product. Write your answer in standard form. (2 points)

5. $(2 - 7i)^2$

$(2 - 7i)(2 - 7i) = 4 - 14i - 14i + 49i^2 = 4 - 28i - 49 = \boxed{-45 - 28i}$

Squaring each part separately loses the $-28i$.

Find the quotient. Write your answer in standard form. (3 points)

6. $\frac{5 - i}{3 + 2i}$

$\frac{5 - i}{3 + 2i} \cdot \frac{3 - 2i}{3 - 2i}$ top: $15 - 10i - 3i + 2i^2 = 13 - 13i$ bottom: $9 + 4 = 13$

$\frac{13 - 13i}{13} = \boxed{1 - i}$

Solve the equation over the set of complex numbers. (4 points)

7. $3x^2 + 5 = 2x$

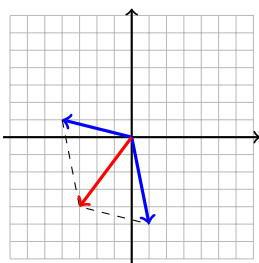
$$3x^2 - 2x + 5 = 0 \quad b^2 - 4ac = 4 - 60 = -56 \quad \sqrt{-56} = 2i\sqrt{14}$$

$$x = \frac{2 \pm 2i\sqrt{14}}{6} = \boxed{\frac{1}{3} \pm \frac{\sqrt{14}}{3}i}$$

Find the sum of the pair of complex numbers. Write your answer in standard form. Graph both complex numbers and their resultant. Then find the absolute value of the sum. (3 points)

8. $-4 + i, 1 - 5i$

$$(-4 + 1) + (1 - 5)i = \boxed{-3 - 4i} \quad |-3 - 4i| = \sqrt{9 + 16} = \boxed{5}$$



Write the complex number in rectangular form. (3 points)

9. $4 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$

$$\frac{5\pi}{6} = 150^\circ: \text{QII, reference } 30^\circ; \cos = -\frac{\sqrt{3}}{2}, \sin = \frac{1}{2}$$

$$4 \left(-\frac{\sqrt{3}}{2} \right) + 4 \left(\frac{1}{2} \right) i = \boxed{-2\sqrt{3} + 2i}$$

Write each complex number in trigonometric (polar) form, with θ in the interval $[0^\circ, 360^\circ)$. (2 points each)

10.

a) $1 - i\sqrt{3}$

b) $-7i$

a) QIV. $r = \sqrt{1 + 3} = 2$

$\tan \theta = -\sqrt{3}$, ref 60°

$\theta = 360^\circ - 60^\circ = 300^\circ$

$$\boxed{2(\cos 300^\circ + i \sin 300^\circ)}$$

b) negative imaginary axis

$r = 7, \theta = 270^\circ$

$$\boxed{7(\cos 270^\circ + i \sin 270^\circ)}$$

The Julia set: compute $z^2 - 1, (z^2 - 1)^2 - 1, [(z^2 - 1)^2 - 1]^2 - 1, \dots$. If the absolute value of any result exceeds 2, z is **not** in the Julia set; otherwise it is. (3 points)

11. Is $z = \frac{1}{2}i$ in the Julia set? Show work.

$$z^2 - 1 = -\frac{1}{4} - 1 = -\frac{5}{4}, \quad \left| -\frac{5}{4} \right| \leq 2 \quad \left(-\frac{5}{4} \right)^2 - 1 = \frac{9}{16} \leq 2 \quad \left(\frac{9}{16} \right)^2 - 1 = -\frac{175}{256}, \quad |\cdot| < 1$$

No result exceeds 2.

YES